Seminari Informal de Matemàtiques de Barcelona

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Date: Wednesday, March 26th, 2025.
Schedule: 13:00 talk.
Place: UB (FMI aula IA) and Zoom.
Language: English.

Title: Automorphism groups of bordered non-orientable Klein surfaces of topological genus 4.

Abstract: We obtain the automorphism groups of compact non-orientable Klein surfaces of topological genus 4 with $k > 0$ boundary components. These groups clearly depend on the number of boundary components, that is, the value of k. The starting point is the analogous result for surfaces without boundary components, that is $k = 0$, obtained by Bujalance, Etayo and Martínez (2014), since it can be proven that a group that acts on a surface with boundary components also acts on surfaces of the same topological type without boundary components. The proof is parallel to the approach followed by Bujalance, Etayo and Martínez (2023) for surfaces of topological genus 3. It is worth noting that as the topological genus of the surface increases, so do the order of the groups of automorphisms and the complexity of their algebraic structure, and hence the difficulty of the computational problem.

About us: *SIMBa* is a mathematics seminar organized by graduate students in the Barcelona area. It is aimed towards graduate and last–year undergraduate students. Our goals are disseminating knowledge from different branches of mathematics for those interested and promoting networking between the attendants.
This seminar is backed by the Faculty of Mathematics and Computer Science at Universitat de Barcelona, Faculty of Mathematics and Statistics at Universitat Politècnica de Catalunya, the Department of Mathematics at Universitat Autònoma de Barcelona, CRM, IMUB and BGSMath.

Fore more information, visit seminari-simba.github.io/en. You may contact us by sending an email to ***seminari.simba@gmail.com***.